

Technology Stack

Date	2 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, Bootstrap	Used to create a responsive and user-friendly interface for entering applicant details and displaying prediction results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask provides a lightweight web framework to connect the machine learning model with the user interface and process prediction requests efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (Application_Data.csv)	Stores applicant information for training and testing the machine learning model. Suitable for this academic project.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render / Localhost (Flask Deployment)	Used to deploy and host the application, making it accessible through a web browser. Localhost is used during development and testing.
5	Version Control & CI/CD	Git, GitHub	Git tracks code changes, while GitHub stores the project online, supports collaboration, and maintains version history.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, Pandas, NumPy, Matplotlib, XGBoost, VS Code	Scikit-learn and XGBoost are used to build and evaluate the prediction model. Pandas and NumPy handle data preprocessing, Matplotlib is used for data visualization, and VS Code is used for development.